

Building a First-to-Default Pricer

From Bootstrapped CDS Curves to a Gaussian-Copula Basket Model

Implementation guide based on the existing CDS curve and pricer workflow

Core inputs: SOFR discount factors, single-name CDS curves, recoveries
Dependence model: One-factor or full-matrix Gaussian copula
Primary engine: Monte Carlo, benchmarked by one-factor quadrature
Target product: First-to-default and general i -th-to-default baskets

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1 Purpose and Scope

This guide explains how to extend an existing single-name CDS curve bootstrap and CDS pricer into a first-to-default (FTD) basket pricer. The existing CDS work already supplies the marginal default distribution for each reference entity. The additional FTD model must introduce a joint dependence structure, determine the first default time and identity, and value the resulting premium and protection legs.

The implementation should be designed in two layers:

1. **Single-name layer:** discount factors, piecewise hazard rates, survival probabilities, default probabilities, premium-leg PV01 and protection-leg valuation.
2. **Basket layer:** correlated default-time generation, first-default selection, basket cash flows, fair FTD spread, correlation calibration and risk.

The intended first production version is a one-factor Gaussian-copula Monte Carlo model with a separate one-dimensional quadrature benchmark. The framework can subsequently be generalised from first-to-default to i -th-to-default baskets.

2 Economic Interpretation of FTD Derivatives

An FTD basket allows an investor to buy or sell protection on the earliest default in a portfolio of reference names. Since the protection seller is exposed to the first default among several names, the fair FTD spread is normally materially above the simple average single-name spread. Market descriptions sometimes characterise this as approximately two to three times the basket-average spread, but this is only a heuristic. The actual multiple depends on:

- the number of names;
- each name's CDS term structure;
- recovery assumptions;
- basket maturity;
- correlation;
- concentration and heterogeneity;
- coupon and accrual conventions.

The FTD spread is generally negatively related to default correlation. When correlation is low, each name is a relatively independent opportunity for the basket to experience at least one default. When correlation is high, the names behave more like a single common credit exposure, reducing the probability that at least one name defaults by a given horizon.

For identical names with maturity default probability p :

$$\mathbb{Q}(\tau_{(1)} \leq T) = 1 - (1 - p)^n$$

under independence, whereas under near-perfect positive dependence the probability approaches p rather than $1 - (1 - p)^n$.

An i -th-to-default basket may contain a small bespoke portfolio or a larger basket, commonly between three and twenty-five names. The FTD is the special case $i = 1$.

3 Existing Single-Name Foundation

3.1 Piecewise-Constant Hazard Rates

For reference name i , assume the CDS bootstrap produces the piecewise-constant hazard function

$$\lambda_i(t) = \begin{cases} \lambda_{i,1}, & 0 < t \leq T_1, \\ \lambda_{i,2}, & T_1 < t \leq T_2, \\ \vdots & \\ \lambda_{i,K}, & T_{K-1} < t \leq T_K. \end{cases}$$

For $t \in (T_{k-1}, T_k]$, the cumulative hazard is

$$\Lambda_i(t) = \sum_{j=1}^{k-1} \lambda_{i,j}(T_j - T_{j-1}) + \lambda_{i,k}(t - T_{k-1}).$$

The risk-neutral survival probability is

$$S_i(t) = \mathbb{Q}(\tau_i > t) = \exp[-\Lambda_i(t)],$$

and the cumulative default probability is

$$F_i(t) = \mathbb{Q}(\tau_i \leq t) = 1 - S_i(t).$$

The bucket hazard $\lambda_{i,k}$ is a conditional instantaneous default intensity. Informally, it represents the rate of default during bucket k , conditional on survival to that point. It is not itself an unconditional bucket default probability.

The conditional probability of surviving from T_{k-1} to T_k is

$$\mathbb{Q}(\tau_i > T_k \mid \tau_i > T_{k-1}) = e^{-\lambda_{i,k}(T_k - T_{k-1})}.$$

Therefore, the conditional probability of default during that bucket is

$$1 - e^{-\lambda_{i,k}(T_k - T_{k-1})}.$$

3.2 What the Existing CDS Pricer Already Provides

The current CDS curve and pricer framework should already contain, either explicitly or implicitly:

- valuation date and settlement conventions;
- SOFR discount factors $D(0, t)$;
- quoted CDS maturities and par spreads;
- recovery assumptions;
- premium schedules and day-count fractions;
- piecewise hazard rates;
- cumulative hazards;
- survival and default probabilities;
- CDS premium-leg risky PV01;
- CDS protection-leg PV;
- repricing errors against quoted CDS spreads.

These are precisely the marginal inputs required by the Gaussian copula. The FTD extension should reuse them rather than recalibrate the single-name curves inside the basket model.

4 FTD Contract Definition

For an n -name basket, define each reference entity's default time as τ_i . The first default time is

$$\tau_{(1)} = \min(\tau_1, \dots, \tau_n).$$

The identity of the first defaulting name is

$$K = \arg \min_{i \in \{1, \dots, n\}} \tau_i.$$

The protection buyer pays the periodic FTD spread s_{FTD} until the earliest of maturity and first default. If name K defaults first before maturity, the protection buyer receives

$$N_K(1 - R_K),$$

where N_K is the contractual notional attached to that name and R_K is its recovery rate. The fair spread solves

$$\text{PV}_{\text{premium}}(s_{\text{FTD}}) = \text{PV}_{\text{protection}}.$$

Since the premium leg is linear in spread,

$$s_{\text{FTD}} = \frac{\text{PV}_{\text{protection}}}{\text{FTD risky annuity}}$$

or, in basis points,

$$s_{\text{FTD}}^{\text{bps}} = 10,000 \frac{\text{PV}_{\text{protection}}}{\text{FTD risky annuity}}.$$

5 Why a Copula Is Required

The bootstrapped CDS curves determine each marginal distribution $F_i(t)$, but they do not determine the joint law

$$\mathbb{Q}(\tau_1 \leq t_1, \dots, \tau_n \leq t_n).$$

A copula joins the marginal distributions into a joint distribution while preserving each marginal curve. Under a Gaussian copula,

$$C_{\Sigma}(u_1, \dots, u_n) = \Phi_{\Sigma}(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_n)),$$

where Φ_{Σ} is the multivariate standard normal CDF with correlation matrix Σ .

The implementation therefore follows the chain

$$\text{CDS spreads} \rightarrow \lambda_i(t) \rightarrow F_i(t) \rightarrow \text{Gaussian copula} \rightarrow (\tau_1, \dots, \tau_n) \rightarrow \tau_{(1)} \rightarrow \text{FTD cash flows}.$$

6 Gaussian-Copula Default-Time Simulation

6.1 Full Correlation-Matrix Form

Let

$$\mathbf{Z} = (Z_1, \dots, Z_n)^{\top} \sim N(\mathbf{0}, \Sigma).$$

Compute a Cholesky factorisation

$$\Sigma = LL^\top,$$

generate independent standard normals

$$\varepsilon \sim N(\mathbf{0}, I),$$

and set

$$\mathbf{Z} = L\varepsilon.$$

Convert each latent normal to a uniform:

$$U_i = \Phi(Z_i).$$

Then convert the uniform into a default time:

$$\tau_i = F_i^{-1}(U_i).$$

This construction preserves the calibrated marginal default distribution:

$$\mathbb{Q}(\tau_i \leq t) = F_i(t).$$

6.2 One-Factor Form

A simpler and highly practical first implementation is

$$Z_i = \beta_i Y + \sqrt{1 - \beta_i^2} \varepsilon_i,$$

where Y and the ε_i are independent standard normal variables. Then

$$\text{Corr}(Z_i, Z_j) = \beta_i \beta_j.$$

In the homogeneous case,

$$\beta_i = \sqrt{\rho}$$

for all i , giving

$$Z_i = \sqrt{\rho} Y + \sqrt{1 - \rho} \varepsilon_i$$

and pairwise latent correlation ρ .

The parameter ρ is a latent Gaussian correlation. It is not directly a CDS-spread correlation, hazard-rate correlation, default-time Pearson correlation or observed equity-return correlation.

6.3 Default-Time Inversion for Piecewise Hazards

Given U_i , define the exponential threshold

$$H_i = -\ln(1 - U_i).$$

Since

$$F_i(t) = 1 - e^{-\Lambda_i(t)},$$

the simulated default time solves

$$\Lambda_i(\tau_i) = H_i.$$

Find bucket k such that

$$\Lambda_i(T_{k-1}) < H_i \leq \Lambda_i(T_k).$$

Then

$$\tau_i = T_{k-1} + \frac{H_i - \Lambda_i(T_{k-1})}{\lambda_{i,k}}$$

provided $\lambda_{i,k} > 0$.

If H_i lies beyond the final calibrated maturity, the model requires an explicit extrapolation rule. The simplest is a flat terminal hazard:

$$\lambda_i(t) = \lambda_{i,K}, \quad t > T_K.$$

7 Monte Carlo Valuation

For path m , generate all default times and identify

$$\tau_{(1)}^{(m)} = \min_i \tau_i^{(m)}, \quad K^{(m)} = \arg \min_i \tau_i^{(m)}.$$

7.1 Protection Leg

The pathwise protection PV is

$$\text{PV}_{\text{prot}}^{(m)} = \mathbf{1}_{\{\tau_{(1)}^{(m)} \leq T\}} D(0, \tau_{(1)}^{(m)}) N_{K^{(m)}} (1 - R_{K^{(m)}}).$$

The Monte Carlo estimate is

$$\text{PV}_{\text{prot}} \approx \frac{1}{M} \sum_{m=1}^M \text{PV}_{\text{prot}}^{(m)}.$$

7.2 Scheduled Premium Leg

For premium dates $t_1, \dots, t_J$ and accrual fractions α_j , the scheduled risky annuity on path m is

$$A_{\text{sched}}^{(m)} = \sum_{j=1}^J \alpha_j D(0, t_j) \mathbf{1}_{\{\tau_{(1)}^{(m)} > t_j\}}.$$

7.3 Accrued Premium on Default

If

$$t_{j-1} < \tau_{(1)}^{(m)} \leq t_j,$$

the accrued contribution is

$$A_{\text{accr}}^{(m)} = D(0, \tau_{(1)}^{(m)}) \text{DCF}(t_{j-1}, \tau_{(1)}^{(m)}).$$

Therefore,

$$A_{\text{FTD}} \approx \frac{1}{M} \sum_{m=1}^M \left(A_{\text{sched}}^{(m)} + A_{\text{accr}}^{(m)} \right).$$

The fair spread is then

$$s_{\text{FTD}} = \frac{\text{PV}_{\text{prot}}}{A_{\text{FTD}}}.$$

8 One-Factor Quadrature Benchmark

Monte Carlo should be the general engine, but the one-factor model also provides a strong independent benchmark.

Define the default threshold

$$a_i(t) = \Phi^{-1}(F_i(t)).$$

Conditional on $Y = y$,

$$Z_i \mid Y = y \sim N(\beta_i y, 1 - \beta_i^2).$$

Hence the conditional default probability is

$$p_i(t \mid y) = \Phi \left(\frac{a_i(t) - \beta_i y}{\sqrt{1 - \beta_i^2}} \right),$$

and conditional survival is

$$q_i(t \mid y) = 1 - p_i(t \mid y).$$

Conditional independence gives

$$\mathbb{Q}(\tau_{(1)} > t \mid Y = y) = \prod_{i=1}^n q_i(t \mid y).$$

Integrating over Y gives the basket survival probability

$$S_{\text{FTD}}(t) = \int_{-\infty}^{\infty} \phi(y) \prod_{i=1}^n \Phi \left(\frac{\beta_i y - \Phi^{-1}(F_i(t))}{\sqrt{1 - \beta_i^2}} \right) dy.$$

This is a one-dimensional integral regardless of basket size and can be evaluated efficiently with Gauss–Hermite quadrature.

For common recovery and notional,

$$\text{PV}_{\text{prot}} = N(1 - R) \int_0^T D(0, t) d[1 - S_{\text{FTD}}(t)].$$

The scheduled annuity is

$$A_{\text{sched}} = \sum_{j=1}^J \alpha_j D(0, t_j) S_{\text{FTD}}(t_j).$$

A simple midpoint approximation for accrual is

$$A_{\text{accr}} \approx \sum_{j=1}^J D(0, \bar{t}_j) \frac{\alpha_j}{2} [S_{\text{FTD}}(t_{j-1}) - S_{\text{FTD}}(t_j)].$$

9 Workbook and Sheet Architecture

9.1 Important Repository Limitation

The public repository confirms the broader curve-bootstrapping project and its SOFR/Excel implementation documentation. However, the binary CDS workbook's internal tabs were not available through the repository connector used for this review. Therefore, the tab names below are a professional target architecture and mapping for the CDS Pricer 2.2 workflow, not a claim that these are the exact existing sheet names.

Where an existing sheet already performs the stated function, retain it and add the specified output columns rather than duplicating it.

9.2 Recommended Existing-Sheet Mapping

Sheet / role	Current CDS function	FTD use and required extension
Control	Valuation date, trade date, settlement date, day-count assumptions, recovery, curve selection and global switches.	Add basket maturity, basket notional, number of names, correlation model, scalar ρ or matrix selector, simulation count, seed, antithetic switch and extrapolation rule.
Discount Curve	Stores SOFR pillar dates, discount factors, zero rates and interpolation inputs.	No conceptual change. Expose a clean function or lookup range returning $D(0, t)$ for arbitrary coupon and simulated default dates.
CDS Market Data	Contains single-name CDS spreads by maturity, recovery assumptions and possibly upfront quotes.	Convert to a basket input table with one row per name and one column per quoted maturity. Add name-specific notional, recovery and sector metadata.
CDS Schedule	Builds coupon dates, accrual fractions, protection start and maturity conventions.	Create one master FTD premium schedule because the basket terminates on first default. Confirm all names use compatible contract dates or define a common basket schedule explicitly.
Hazard Bootstrap	Solves sequentially for piecewise hazards that reprice each CDS quote.	Run independently for every basket name. Output a rectangular table indexed by name and hazard bucket.
Survival Curve	Calculates cumulative hazard, survival probability and default probability at curve nodes or coupon dates.	Expose $F_i(t)$ and $F_i^{-1}(u)$. Add cumulative-hazard breakpoint arrays used by default-time inversion.
CDS Pricer	Calculates premium leg, accrued premium, protection leg, par spread and PV for one name.	Use as a single-name validation engine. A one-name FTD must reproduce this sheet's par CDS spread under identical conventions.
Calibration Check	Shows quote repricing errors and curve monotonicity.	Add basket-wide pass/fail flags: non-negative hazards, decreasing survival, marginal simulation checks and maximum repricing error.

9.3 New FTD Sheets to Add

New sheet	Purpose	Principal contents
FTD Basket Input	Defines the product and constituent-level terms.	Name, weight/notional, recovery, CDS curve reference, maturity, currency, seniority, correlation loading β_i , inclusion flag.
FTD Correlation	Stores and validates the dependence model.	Scalar ρ , one-factor loadings, or full matrix Σ ; symmetry checks; diagonal checks; eigenvalues; Cholesky factor.
FTD Randoms	Generates systemic and idiosyncratic random variables.	$Y, \varepsilon_i, Z_i, U_i$ by simulation path. Prefer an external Python engine for large path counts.
FTD Default Times	Maps copula uniforms through each marginal curve.	$H_i = -\ln(1 - U_i)$, selected hazard bucket, inverted τ_i , first default time and first default identity.
FTD Cash Flows	Calculates pathwise premium and protection legs.	Scheduled survival indicators, accrued premium, loss-given-default, discount factors and pathwise PVs.
FTD Results	Reports price and fair spread.	Protection PV, risky annuity, fair spread, standard error, confidence interval, expected first-default time and first-name contribution.
FTD Sensitivities	Calculates risk and scenario analysis.	Correlation01, constituent CS01, recovery sensitivity, IR01, basket-name contribution and correlation grid.
FTD Validation	Runs theoretical and numerical test cases.	One-name equivalence, zero-correlation identity, near-one-correlation limit, marginal preservation and Monte Carlo versus quadrature.

10 Detailed Excel Implementation

10.1 Step 1: Standardise the Name-by-Bucket Hazard Table

Create a table with one row per name and two aligned blocks:

1. hazard rates $\lambda_{i,k}$;
2. cumulative hazards $\Lambda_i(T_k)$.

For name i and bucket k :

$$\Lambda_i(T_k) = \Lambda_i(T_{k-1}) + \lambda_{i,k}(T_k - T_{k-1}).$$

In Excel, conceptually:

```
=CumulativeHazard_Previous
+ Hazard_Current * (BucketEnd - BucketStart)
```

Ensure all time differences use the same year-fraction convention as the bootstrap.

10.2 Step 2: Add Inverse Default-Time Logic

For each path and name:

```
U = NORM.S.DIST(Z, TRUE)
H = -LN(1-U)
```

Use `XMATCH` or `MATCH` to find the first cumulative-hazard breakpoint greater than or equal to H . Conceptually:

```
Bucket = XMATCH(H, CumulativeHazardRow, 1)
Tau = BucketStart
      + (H - CumHazardAtBucketStart) / HazardInBucket
```

Care is required because approximate-match modes differ across Excel versions. Test boundary values exactly equal to each cumulative-hazard breakpoint.

10.3 Step 3: Generate Correlated Normals

For a homogeneous one-factor model:

```
SystemicY = NORM.S.INV(RAND())
IdioEps = NORM.S.INV(RAND())
Z_i = SQRT(rho)*SystemicY + SQRT(1-rho)*IdioEps
```

For reproducibility, ordinary volatile `RAND()` formulas are not ideal. A Python engine should generate and save a fixed random-number matrix using a specified seed.

10.4 Step 4: Select the First Default

For each path:

```
FirstDefaultTime = MIN(DefaultTime_Name1:DefaultTime_NameN)
FirstNameIndex = XMATCH(FirstDefaultTime,
                        DefaultTime_Name1:DefaultTime_NameN,
                        0)
```

Simultaneous defaults have zero probability under a continuous Gaussian-copula model. Nevertheless, implement a deterministic tie rule for numerical robustness.

10.5 Step 5: Calculate Pathwise Protection PV

```
=IF(FirstDefaultTime<=Maturity,
    DF_at_FirstDefault
    * Notional_of_FirstName
    * (1-Recovery_of_FirstName),
    0)
```

The discount factor at a simulated date must be interpolated from the SOFR curve using the same interpolation convention as the existing pricer.

10.6 Step 6: Calculate Scheduled Premium PV01

For coupon date t_j :

```
=AccrualFactor_j
* DF_j
* --(FirstDefaultTime > CouponDate_j)
```

Sum across all coupon dates.

10.7 Step 7: Calculate Accrued Premium

Find the coupon interval containing the first default time. Then calculate

```
=IF(FirstDefaultTime<=Maturity,
    DF_at_FirstDefault
    * YearFrac(PreviousCouponDate, FirstDefaultTime),
    0)
```

Use the contractual accrual convention, not a simple calendar-day fraction unless that is exactly the convention in the current CDS pricer.

10.8 Step 8: Aggregate and Solve Fair Spread

```
ProtectionPV = AVERAGE(PathwiseProtectionPV)
RiskyAnnuity = AVERAGE(PathwiseScheduledPV01
                        + PathwiseAccruedPV01)
FairSpread = ProtectionPV / RiskyAnnuity
FairSpreadBp = 10000 * FairSpread
```

Also report the Monte Carlo standard error:

$$SE(\bar{X}) = \frac{s_X}{\sqrt{M}}.$$

For the ratio defining the fair spread, use either common-path bumping or a delta-method estimate if a spread confidence interval is required.

11 Recommended Python Calculation Engine

Excel is appropriate for transparency, controls and reporting. The simulation itself should preferably be implemented in Python and returned to Excel as summarised outputs.

11.1 Core Data Structures

```
@dataclass
class HazardCurve:
    breakpoints: np.ndarray
    hazards: np.ndarray
    cumulative_hazards: np.ndarray
    recovery: float

@dataclass
class FTDBasket:
    names: list[str]
    notionals: np.ndarray
    curves: list[HazardCurve]
    maturity: float
    premium_dates: np.ndarray
    accrual_fractions: np.ndarray
```

11.2 Default-Time Inversion

```
def inverse_default_times(u, breakpoints, hazards, cumulative):
    h = -np.log1p(-u)
    k = np.searchsorted(cumulative, h, side="left")
```

```

tau = np.empty_like(h)

inside = k < len(hazards)
k_in = k[inside]

previous_cum = np.where(
    k_in == 0, 0.0, cumulative[k_in - 1]
)
previous_time = np.where(
    k_in == 0, 0.0, breakpoints[k_in - 1]
)

tau[inside] = (
    previous_time
    + (h[inside] - previous_cum) / hazards[k_in]
)

beyond = ~inside
tau[beyond] = (
    breakpoints[-1]
    + (h[beyond] - cumulative[-1]) / hazards[-1]
)

return tau

```

11.3 One-Factor Simulation

```

rng = np.random.default_rng(seed)

y = rng.standard_normal((n_paths, 1))
eps = rng.standard_normal((n_paths, n_names))

z = beta.reshape(1, -1) * y + np.sqrt(
    1.0 - beta.reshape(1, -1) ** 2
) * eps

u = scipy.stats.norm.cdf(z)

```

Use vectorised inversion by name, obtain the row-wise minimum, and calculate pathwise cash flows.

12 Correlation Input and Calibration

Single-name CDS quotes do not determine correlation. Correlation must come from an additional market input or scenario assumption.

12.1 Implied FTD Correlation

If a market FTD spread is available, solve

$$s_{\text{model}}(\rho) - s_{\text{market}} = 0.$$

Use bisection or Brent's method on a bounded interval such as

$$\rho \in [0, 0.999].$$

The model spread should generally decline with ρ for an FTD basket, but verify monotonicity numerically for heterogeneous portfolios and contractual features.

12.2 Alternative Inputs

Possible alternatives include:

- historical equity or asset-return correlations;
- sector-factor models;
- index-tranche-implied dependence;
- a conservative scenario grid.

Historical correlations are physical-measure estimates and are not automatically risk-neutral. They should be labelled as assumptions rather than market calibrations.

13 Risk Measures

The minimum risk report should include:

Correlation01: Change in PV or fair spread for a one-percentage-point change in ρ .

Constituent CS01: Change in basket PV after bumping one name's CDS spread curve by one basis point and re-bootstrapping that name.

Parallel basket CS01: Simultaneous one-basis-point bump to all constituent CDS curves.

Recovery sensitivity: Change in PV after changing one or all recovery assumptions.

IR01: Change in PV after a one-basis-point shift to the discount curve.

Name contribution: Expected discounted loss contribution by first-defaulting name.

Use common random numbers for all bumped Monte Carlo calculations. This materially reduces noise in finite-difference sensitivities.

14 Validation Framework

14.1 Single-Name Equivalence

For $n = 1$, the FTD fair spread must equal the existing single-name CDS par spread, subject to identical timing, accrual, discounting and recovery conventions.

14.2 Zero-Correlation Check

At $\rho = 0$,

$$S_{\text{FTD}}(t) = \prod_{i=1}^n S_i(t).$$

The Monte Carlo result should converge to this analytical identity.

14.3 Perfect-Dependence Limit

For identical names and $\rho \rightarrow 1$, the FTD should approach a single-name CDS because all latent variables and default times converge.

14.4 Marginal Preservation

For each name and date t ,

$$\frac{1}{M} \sum_{m=1}^M \mathbf{1}_{\{\tau_i^{(m)} \leq t\}} \approx F_i(t).$$

Changing correlation must not alter the unconditional marginal default curve.

14.5 Monte Carlo versus Quadrature

Under the one-factor model with common recovery and notional, Monte Carlo and Gauss–Hermite quadrature should agree within Monte Carlo sampling error.

14.6 Correlation-Matrix Checks

For a full matrix model, require:

$$\Sigma = \Sigma^\top, \quad \Sigma_{ii} = 1, \quad \Sigma \succeq 0.$$

If the matrix is not positive semidefinite, Cholesky decomposition will fail or produce an invalid dependence structure.

15 Extension to i -th-to-Default

The simulation framework generalises directly. Sort each path’s default times:

$$\tau_{(1)} \leq \tau_{(2)} \leq \dots \leq \tau_{(n)}.$$

For an i -th-to-default contract, termination and protection payment occur at $\tau_{(i)}$ rather than $\tau_{(1)}$.

In Python:

```
ith_time = np.partition(default_times, i - 1, axis=1)[: , i - 1]
ith_name = np.argpartition(default_times, i - 1, axis=1)[: , i - 1]
```

The premium leg survives until the i -th default. The protection payment uses the recovery and notional of the i -th defaulting name. For $i > 1$, correlation sensitivity can differ substantially from the FTD case because the product requires multiple defaults rather than merely one.

16 Recommended Build Order

1. Freeze and validate the existing SOFR discount curve.
2. Validate each single-name hazard curve by repricing its quoted CDS spreads.
3. Refactor the hazard output into a standard name-by-bucket table.
4. Implement and unit-test inverse default-time mapping.
5. Build a homogeneous one-factor Gaussian-copula simulator.
6. Add first-default time and identity selection.
7. Add scheduled premium, accrued premium and protection cash flows.
8. Produce fair spread, PV and Monte Carlo error.
9. Implement the one-factor quadrature benchmark.

10. Add correlation scenarios and implied-correlation calibration.
11. Add constituent CS01, correlation01, recovery risk and IR01.
12. Generalise from first-to-default to i -th-to-default.

17 Model Limitations

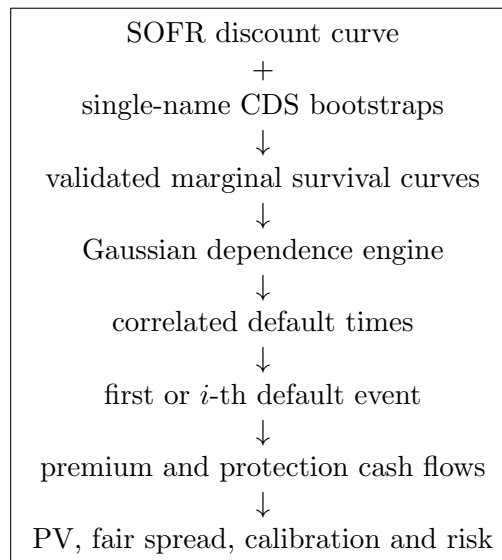
The Gaussian copula is useful because it is tractable and preserves calibrated single-name marginals. Its limitations must remain visible:

- a constant latent correlation does not capture correlation skew;
- the Gaussian copula has no lower-tail dependence beyond that implied by linear Gaussian correlation;
- static default-time copulas do not model spread dynamics;
- contagion after a realised default is not represented;
- historical correlations may differ materially from risk-neutral dependence;
- recovery is often treated as deterministic even though recovery and default dependence may be material;
- bespoke-basket correlation calibration may be underdetermined.

The model should therefore report correlation scenarios and concentration measures rather than presenting one correlation assumption as uniquely correct.

18 Final Target Architecture

The completed system should have the following separation:



The existing CDS Pricer 2.2 should remain the authoritative single-name calibration and validation layer. The FTD module should consume its hazard and survival outputs, not recreate them independently. This gives a transparent lineage from market CDS spreads to each constituent's marginal curve and then to the basket valuation.

References

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